Demo Version of the Second-Round Tasks

Computer and Data Science: Second-Round Sample Tasks for the Open Doors Master's and Doctoral Tracks

This sample test includes 46 tasks: 25 basic-level test tasks with one correct answer, 15 intermediate-level tasks with several correct answers or a model answer, 3 advanced-level tasks requiring a detailed answer, and 3 oral-response tasks. For tasks requiring a detailed answer and for oral-response tasks, assessment criteria and model answers are provided.

Task 1

Basic level (maximum 1 point)

Field of science: Applied mathematics

As x approaches what value does the function $y=x^2+4x+4$ become infinitesimal?

a) -2

b) 3

c) 4

d) 5

Answer: a

Task 2

Basic level (maximum 1 point)

Field of science: Applied mathematics

To which type does this series belong?

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

a) Alternating series

b) Power series

c) Fourier series

d) A series with positive terms

Answer: a

Task 3

Basic level (maximum 1 point)

Field of science: Applied mathematics

Specify the type of integral:

$$\int \frac{1}{x}\,dx = \ln|x| + C, \quad x \neq 0.$$

a) indefinite integral

b) definite integral

c) improper integral

d) line integral

Answer: a

Task 4

Basic level (maximum 1 point)

Field of science: Computer science, information systems

How many bits make up one nibble (half-byte)?

a) 2

b) 3

c) 4

d) 6

Answer: c

Task 5

Basic level (maximum 1 point)

Field of science: Computer science, information systems

Which device is used for long-term data storage and retains data even when the power is turned off?

a) Random Access Memory (RAM)

b) CPU cache memory

c) Hard Disk Drive (HDD)

d) Video RAM (VRAM)

Answer: c

Task 6

Basic level (maximum 1 point)

Field of science: Computer science, information systems

Which of the following is an example of information protection?

a) Collecting user feedback about a new app

b) Generating a quarterly sales chart

c) Encrypting a laptop's hard drive contents

d) Publishing news on a corporate portal

Answer: c

Task 7

Basic level (maximum 1 point)

Field of science: Computer science, information systems

What is the main goal of a data backup system?

a) Accelerating the performance of company servers

b) Protecting computers from viruses and malware

c) Creating copies of critical data for recovery after failures or losses

d) Organizing video conferences between employees

Answer: c

Task 8

Basic level (maximum 1 point)

Field of science: Computer science, information systems

What does the acronym RAID stand for in the context of data storage systems?

a) Rapid Access to Information Data

b) Random Array of Independent Disks

c) Redundant Array of Independent Disks

d) Reliable Access for Internet Data

Answer: c

Task 9

Basic level (maximum 1 point)

Field of science: Computer science, artificial intelligence

Imagine you are training a model that achieves high accuracy on training data but performs significantly worse on new data. What phenomenon does this situation illustrate?

a) Underfitting

b) Overfitting

c) Bias-Variance Trade-off

d) Stability

Answer: b

Task 10

Basic level (maximum 1 point)

Field of science: Computer science, artificial intelligence

You are evaluating a model predicting a rare disease, and it is crucial to minimize the number of missed cases (false negatives). Which metric will be the most critical for monitoring?

a) Accuracy

b) Recall

c) Precision

d) F1-score

Answer: b

Task 11

Basic level (maximum 1 point)

Field of science: Computer science, artificial intelligence

Which of the two regression metrics is more sensitive to outliers in the data: Mean Squared Error (MSE) or Mean Absolute Error (MAE)? Why?

a) MSE, because it uses the squares of errors, which increases the influence of large errors

b) MAE, because it uses the squares of errors

c) Both metrics are equally sensitive to outliers

d) Neither of them is sensitive to outliers

Answer: a

Task 12

Basic level (maximum 1 point)

Field of science: Computer science, artificial intelligence

Which clustering algorithm can detect clusters of arbitrary shapes and handles noise well, without requiring the number of clusters to be specified in advance?

a) k-Means

b) Hierarchical clustering

c) DBSCAN

d) EM algorithm

Answer: c

Task 13

Basic level (maximum 1 point)

Field of science: Computer science, interdisciplinary applications

What category of SQL commands do COMMIT and ROLLBACK belong to?

a) TCL (Transaction Control Language)

b) DCL (Data Control Language)

c) DML (Data Manipulation Language)

d) DDL (Data Definition Language)

Answer: a

Task 14

Basic level (maximum 1 point)

Field of science: Computer science, interdisciplinary applications

Which of the following are deterministic interpolation methods used in geoinformatics?

a) Universal Kriging

b) Simple Kriging

c) Cogenetic variant

d) IDW (Inverse Distance Weighting)

Answer: d

Task 15

Basic level (maximum 1 point)

Field of science: Computer science, interdisciplinary applications

Which component of a Decision Support System (DSS) is responsible for mathematical models, “what-if” scenarios, and optimization?

a) Model Management Component

b) Data Management Component

c) User Interface Component

d) Knowledge Base

Answer: a

Task 16

Basic level (maximum 1 point)

Field of science: Computer science, software engineering

Which of the following is a data type?

a) Array

b) Integer

c) String

d) All of the above

Answer: d

Task 17

Basic level (maximum 1 point)

Field of science: Computer science, software engineering

What is a development phase?

a) The output of a specific stage of a process

b) A specific type of work performed during the software development process

c) The structure according to which software development is built

d) A specific stage of a process that has a beginning, an end, and an output

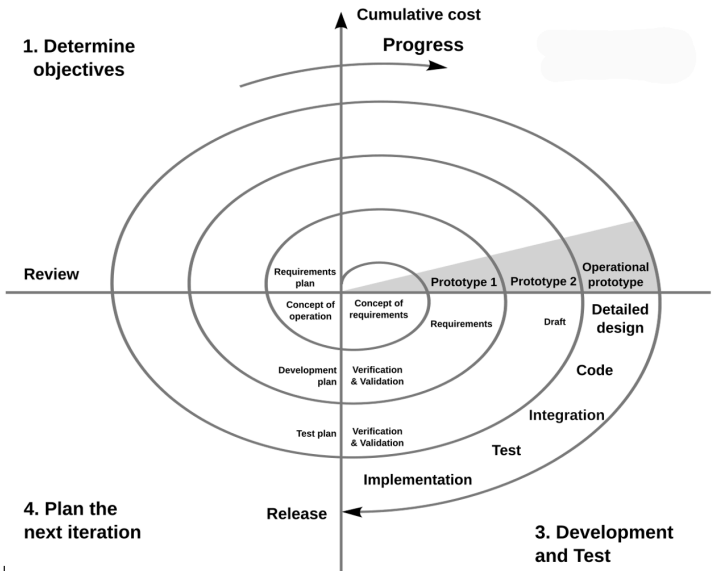
Answer: d

Task 18

Basic level (maximum 1 point)

Field of science: Computer science, software engineering

The spiral model of software engineering (Boehm and Papaccio, 1988) has four major processes, starting in the top left of the figure. Which process is the second one?



a) Identify and resolve risks

b) Identification and evaluation of alternative approaches

c) Development and testing

d) Planning for the next iteration

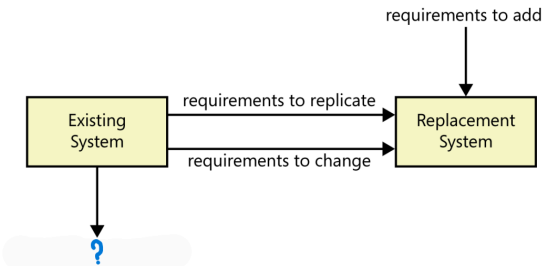
Answer: a

Task 19

Basic level (maximum 1 point)

Field of science: Computer science, software engineering

When performing a gap analysis for an information system replacement project, four types of requirements are considered. Which requirement type should replace the question mark in the diagram?



a) Requirements to be modified

b) Requirements to be drop

c) Requirements to be clarified

d) Requirements allocated to a separate module

Answer: b

Task 20

Basic level (maximum 1 point)

Field of science: Computer science, hardware & architecture

A processor with a direct-mapped cache has 4 GB of main memory, a cache line size of 64 bytes, and 1024 cache lines. How many bits are used to select the cache line (index)?

a) 16

b) 32

c) 4

d) 10

Answer: d

Task 21

Basic level (maximum 1 point)

Field of science: Computer science, hardware & architecture

What type of cache miss occurs when the processor accesses data for the first time?

a) Compulsory misses

b) Capacity misses

c) Conflict misses

d) Coherence-related errors

Answer: a

Task 22

Basic level (maximum 1 point)

Field of science: Computer science, hardware & architecture

A device transmits data over USB 2.0 at approximately 80% of the theoretical speed. How many megabytes (MB) can be transferred in 20 seconds?

a) 1240

b) 720

c) 960

d) 1200

Answer: c

Task 23

Basic level (maximum 1 point)

Field of science: Computer science, hardware & architecture

What is a "zombie process" in Unix-like operating systems?

a) A process that consumes CPU time indefinitely

b) A terminated process whose entry remains in the process table until the parent process retrieves its exit status code

c) A process blocked due to memory shortage

d) A process that cannot be terminated by the `kill` command

Answer: b

Task 24

Basic level (maximum 1 point)

Field of science: Computer science, hardware & architecture

In which cases may the scheduler change a thread's dynamic priority?

a) When its allocated time quantum expires or when the thread is unblocked

b) Only when the thread is launched

c) Only upon abnormal termination of the thread

d) Never — priorities are fixed

Answer: a

Task 25

Basic level (maximum 1 point)

Field of science: Computer science, hardware & architecture

What is the purpose of a page table in paged memory management?

a) To speed up process execution

b) To implement swapping

c) To translate (map) a virtual address into a physical address

d) For process synchronization

Answer: c

Task 26

Intermediate level (maximum 3 points)

Field of science: Applied mathematics

How many vertical asymptotes does the graph of the function have?

$$f(x) = \frac{x}{x^4 - 16}$$

Answer: 2

Task 27

Intermediate level (maximum 3 points)

Field of science: Applied mathematics

Suppose *f*(*x*) is a twice-differentiable function on the open interval (*a*,*b*). What can be said about the behavior of *f*(*x*) on (*a*,*b*)?

$$f'(x_0) > 0, \forall x_0 \in (a, b)$$

Answer: It is increasing

Task 28

Intermediate level (maximum 3 points)

Field of science: Applied mathematics

Arrange the integrals in ascending order of their values.

$$1) \int_{-\pi}^{\pi} \sin x \ dx$$

$$2) \int_{0}^{\frac{\pi}{2}} \sin^2 x \cdot \cos x \ dx$$

$$3) \int_{0}^{\frac{\pi}{2}} \sin x \cdot \cos x \ dx$$

$$4) \frac{1}{\pi} \int_{-\pi}^{\pi} \sin^2 x \ dx$$

(Write the numbers without spaces, brackets, or commas.)

Answer: 1234

Task 29

Intermediate level (maximum 3 points)

Field of science: Computer science, information systems

During the design of the `notes` table in a relational DBMS, the `content` field has the data type `VARCHAR(250)` and is stored using ASCII encoding (1 byte per character). The field is fully populated with 250 characters. How many kilobytes (KB) does this note occupy in memory? Round the answer to one decimal place (X.X).

Answer: 0.2

Task 30

Intermediate level (maximum 3 points)

Field of science: Computer science, information systems

What is the name of the function that is automatically invoked when an object is destroyed?

Answer: destructor

Task 31

Intermediate level (maximum 3 points)

Field of science: Computer science, artificial intelligence

You are analyzing a time series of sales data for 1 year (365 days). If you apply a moving average with a window of 7 days to smooth the data, how many data points will be lost at the beginning and end of the series due to the inability to compute the average for them (without padding)? Provide the answer as an integer number.

Answer: 6

Task 32

Intermediate level (maximum 3 points)

Field of science: Computer science, artificial intelligence

Which tasks are classic examples of Reinforcement Learning, where an agent learns to make decisions in an environment to maximize reward? Select all that apply.

a) Teaching a program to play Go or chess

b) Spam detection in email

c) Teaching a robot manipulator to pick up and move objects

d) Determining a client's credit score

e) Optimizing navigation for an autonomous vehicle

f) Clustering customers of an online store

Answer: a, c, e

Task 33

Intermediate level (maximum 3 points)

Field of science: Computer science, software engineering

Which of the following issues are addressed in the field of software engineering? Select all that apply.

a) Social policy issues

b) Development team management issues

c) Software development life cycle support issues

d) No correct answer

Answer: b, c

Task 34

Intermediate level (maximum 3 points)

Field of science: Computer science, software engineering

In event-driven architecture for scalable and high-performance applications, when a high level of responsiveness is required, a topology is used where there is no central event mediator and the message flow is distributed across the event processor components in a chain-like broadcasting fashion. What is this topology called? (write down 2 words)

Answer: broker topology

Task 35

Intermediate level (maximum 3 points)

Field of science: Computer science, hardware & architecture

Four physical hard drives are combined into a single logical RAID 5 array. Each drive has a capacity of 1 TB. One drive fails. What is the amount of available and safely usable space in the array at this time? Enter only the integer number of terabytes.

Answer: 2

Task 36

Intermediate level (maximum 3 points)

Field of science: Computer science, hardware & architecture

A host (physical or virtual computer) has 16 GB of RAM. 4 Docker containers are running on the host, each with a memory limit of 3 GB. How much RAM remains available for the host OS if all containers are using their maximum allowed memory? Enter only the integer number of gigabytes.

Answer: 4

Task 37

Intermediate level (maximum 3 points)

Field of science: Computer science, hardware & architecture

Physical memory is divided into frames. Each entry in the page table takes 8 bytes, and the entire table is 16 KB. What is the size of each frame if the system can address a maximum physical memory of 4 MB? Enter the answer in integer kilobytes.

Answer: 2

Task 38

Intermediate level (maximum 3 points)

Field of science: Computer science, hardware & architecture

Which types of data can be stored in NTFS Alternate Data Streams (ADS)? Select all that apply.

a) File metadata (e.g., mark-of-the-web / origin information).

b) Antivirus/virus signatures

c) Image thumbnails and other auxiliary information

d) Executable code that may be hidden from standard file browsing tools

e) File backups

f) Operating system event logs

Answer: a, c, d

Task 39

Intermediate level (maximum 3 points)

Field of science: Computer science, hardware & architecture

Which NTFS technologies are used for integration with Active Directory? Select all that apply.

a) Access Control Lists (ACLs) with support for user and group Security Identifiers (SIDs)

b) Encrypting File System (EFS) tied to user accounts

c) Built-in backup via System Center

d) Disk space quotas with reporting for administrators

e) Automatic encryption of all domain files

Answer: a, b, d

Task 40

Intermediate level (maximum 3 points)

Field of science: Computer science, hardware & architecture

What are the advantages of dynamic resource allocation over static allocation? Select all that apply.

a) Guaranteed allocation of the maximum amount of resources to each process

b) More efficient resource utilization due to flexibility

c) Reduced overhead for resource management

d) Ability to adapt to changing workload

e) Simplified programming due to the lack of need to free resources

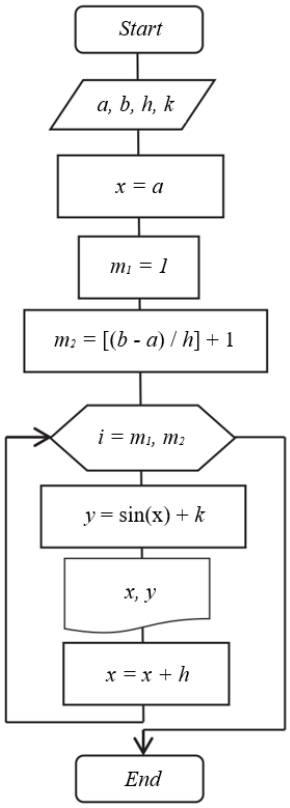
f) Reduction in the number of processes in the system

Answer: b, d

Task 41

Advanced level (maximum 10 points)

Field of science: Computer science, information systems



1. What type of algorithm is represented in the figure?
2. Formulate the condition of the applied task that the program solves, written according to the algorithm shown in the figure.
3. Indicate the most suitable data type used in a high-level programming language for the variables employed in the task. Justify the choice of type by specifying the physical meaning and the permissible range of values.

Solution method: 1.The algorithm belongs to the cyclic computational type with a counter. By control structure, it is a linear-cyclic algorithm: after the input of initial data, a loop body follows, repeating a predetermined number of times. By purpose, the algorithm solves the problem of function tabulation — constructing a table of its values on a fixed interval with a given step.
2.Condition of the applied task for the HR department: the organization's HR department carries out forecasting of the total volume of HR work for the planning period. Based on the statistical analysis of data from previous years, it has been established that the total labor intensity of HR processes y (in conditional units) depends on time x (in years) and is described by the formula: $y=\cos(x)+e^{x/2}$ where $\cos(x)$ reflects the cyclic fluctuations in workload associated with the seasonality of HR activities (assessments, mass hiring,

natural attrition), and $ex/2$ reflects the trend growth of labor intensity following the expansion of staff and business scaling.
Required: with step h , calculate and output a table of forecasted values of HR workload volume on the horizon from a to b years. The obtained data are used to justify the headcount of the HR department, plan the recruiting budget, and develop the staff training schedule.
3.Data type justification: time and labor intensity require fractional representation and high precision, while loop counters and the number of tabulation points must be integer.

Answer:
Algorithm type: cyclic computational algorithm with a parameter, linear-cyclic structure, function tabulation.
Task condition: the HR department forecasts HR process labor intensity using $y=\cos(x)+ex/2$ on $[a;b]$ with step h ; the result is a value table for headcount and budget planning.
Data types:

Variable	Data type	Physical meaning and range	Justification
a , b	Integer (`int`)	Start and end of the planning period (years), integer values, $a\geq 0$, $b>a$	The period is specified by whole year numbers (e.g., year 1, year 5); fractional increment within the loop is handled by step h and variable x
h	Real (float / double)	Discretization step (years), $h>0$	Step may be fractional: quarter (0.25), half-year (0.5)
x	Real (float / double)	Current time in loop (years), $x\in[a;b]$	Changes by fractional step, requires precision
y	Real (double)	Forecasted workload (cond. units), $y>0$	Result of transcendental functions; for $x\geq 20$ the value $ex/2$ requires enhanced precision
$m1$	Integer (int)	Loop counter initial value, $m1=1$	Ordinal number, positive integer
$m2$	Integer (int)	Loop counter final value, $m2=\lfloor (b-a)/h \rfloor +1$	Number of tabulation points is always an integer
i	Integer (int)	Loop parameter, $i\in[m1;m2]$	Iteration counter, integer by definition

Assessment criteria

Criterion 1: The algorithm type is correctly identified and justified — 0–2 points

Criterion 2: A complete and correct applied formulation of the HR department task is provided — 0–3 points

Criterion 3: Correct data types with justification are specified for all variables — 0–3 points

Criterion 4: The physical meaning and permissible range of values are indicated for each variable — 0–2 points

Task 42

Advanced level (maximum 10 points)

Field of science: Computer science, information systems

A code fragment is given:

```
int t=0, i, j, k; i = 2; j = 9;

k=i<j?i++:++j?t=i+j, printf("i=%i\n", i) : i;

cout<<"i="<<i<<endl;

cout<<"j="<<j<<endl;

cout<<"t="<<t<<endl;

cout<<"k="<<k<<endl;
```

1. What operation and in which programming language does this example represent?

2. Provide the code for the same operation in Python, correcting it so that the branch

```
++j ? t = i + j, printf("i=%i\n", i) : i
```

is executed.

3. State the result of the operation execution.

4. The algorithm for automated processing of a student assignments table must analyze each table object in a loop. For each element in the table, a data correctness check is performed (for example, the value is not empty, > 0). For each correct element, a processing branch is selected based on a discrete feature (type, code, category). For each element in a specific branch, the final value is calculated based on a numerical condition. The result of the algorithm's operation is the formation of a resulting collection (a list of categories and a list of statuses). Which branching operators are more appropriate to use for each action?

Solution method: 1. Language: C++
Operation: nested (cascading) ternary conditional operator ?: combined with the comma operator.
The line
k = i < j ? i++ : ++j ? t = i + j, printf("i=%i\n", i) : i;
is read taking into account the right-associativity of ?: as:
k = i < j ? (i++) : ((++j) ? (t = i + j, printf("i=%i\n", i)) : i);

2. For the branch ++j ? t = i + j, printf(...) : i to execute, the condition i < j must be false (i >= j).
Python has no prefix increment ++j and no comma operator in the C++ sense, so the code can look like this:

```
import sys
i, j, t = 10, 9, 0
def printf(fmt, val):
    s = fmt % val
    sys.stdout.write(s)
    return len(s)
if i < j:
    k = i
    i += 1
else:
    j += 1
    if j:
        t = i + j
        k = printf("i=%i\n", i)
    else:
        k = i
```

3. i=10, j=10, t=20, k=5

4. Data correctness check — if–then–else
A compound Boolean condition is evaluated (not empty and > 0). It has exactly two outcomes: "data is correct" and "data is incorrect."
Branch selection by discrete feature — switch
Type, code, or category is a finite set of discrete values. The switch statement explicitly shows that the branches are equal in status and mutually exclusive.
Numerical condition inside a branch — if–then–else or ternary operator
Inside the branch, a numerical inequality or range membership is checked. if–then–else is universal for any comparison. If both branches only assign different values to a single variable, the ternary operator is acceptable.

Answer: 1. C++, operation: ternary operator

2.

```
import sys
i, j, t = 10, 9, 0
def printf(fmt, val):
    s = fmt % val
    sys.stdout.write(s)
```

```
    return len(s)
if i < j:
    k = i
    i += 1
else:
    j += 1
    if j:
        t = i + j
        k = printf("i=%i\n", i)
    else:
        k = i
```

3. i=10, j=10, t=20, k=5

4. if–then–else, switch, if–then–else or ternary operator

Assessment criteria

Criterion 1: The programming language is correctly identified, and the choice is justified. The type of conditional operator is correctly named, accounting for its nesting and syntactic specifics. — 0–2 points

Criterion 2: The code is syntactically correct and executes without errors. The nested branching logic of the original fragment is preserved. — 0–3 points

Criterion 3: Final values of all variables are correctly determined with a step-by-step explanation of the evaluation order. — 0–2 points

Criterion 4: An appropriate branching operator type is selected for each algorithm stage. Brief justification is provided, noting the limitations of alternative choices. — 0–3 points

Task 43

Advanced level (maximum 10 points)

Field of science: Computer science, artificial intelligence



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Fuzzy logic system based on Mamdani algorithm is used to control the air conditioning system.

The system has two inputs:

- Air temperature is a linguistic variable "Temperature" with the values "low", "average", "high".
- Air humidity is a linguistic variable "Humidity" with the meanings "dry", "normal", "wet".

and one output:

- Fan speed – linguistic variable “Speed” with the values “slow”, “medium”, “fast.

Membership functions are given by triangular distributions:

- "Temperature": "low" (0, 0, 20), "medium" (15, 25, 35), "high" (30, 40, 40).
- "Humidity": "dry" (0, 0, 40), "normal" (30, 50, 70), "wet" (60, 100, 100).
- "Speed ": "slow" (0, 0, 50), "medium" (30, 50, 70), "fast" (60, 100, 100)

Rule base:

1. If "Temperature" = "low" and "Humidity" = "dry", then "Speed" = "slow"
2. If "Temperature" = "medium" and "Humidity" = "normal", then "Speed" = "medium"
3. If "Temperature" = "high" and "Humidity" = "wet", then "Speed" = "fast"
4. If "Temperature" = "high" and "Humidity" = "normal", then "Speed" = "medium"

Input values:

- **Temperature = 33 °C**
- **Humidity = 65%**

Task:

1. Write down the formula for the triangular membership function $\mu(x)$ for a point x and a term with a triple (a, b, c)
2. Using the triangular membership function formula, calculate the degrees of membership of the input values to each term.
3. Determine the degrees of activation α_k of output terms for each of the *k* fuzzy rules.
4. Perform output aggregation.
5. Perform center of gravity defuzzification and record the crisp *FanSpeed* value as a percentage (round to whole numbers, discarding the fractional part).
6. Using the interquartile range (IQR) method, identify outlier values in the “Temperature” and “Humidity” parameters (in the table). Record the day numbers in which anomalies are detected.
7. Determine which type of machine learning task this problem belongs to.

Please note: the assessment will take into account the progress of the solution; writing only the answer is not enough.

Solution method: ***Solution:***

1. Let's write the formula for the triangular membership function $\mu(x)$ for a point x and a term with a triple (a, b, c).
2. Using the triangular membership function formula, we calculate the degrees of membership of the input values to each term.
3. Let's determine the degrees of activation α_k of output terms for each of the *k* fuzzy rules.
4. Let's perform output aggregation.
5. Let's perform defuzzification using the center of gravity method and write down the crisp value of the fan speed *FanSpeed* as a percentage (rounded to whole numbers, discarding the fractional part).

$$\mu(x) = \begin{cases} 0, & \leq a \text{ or } x \geq c \\ \dfrac{x-a}{b-a}, & a < x \leq b \\ \dfrac{c-x}{c-b}, & b < x < c \end{cases}$$

Temperature = $33°C$:

"low" (0, 0, 20): $\mu_{low}(33) = 0$

"average" (15, 25, 35): $\mu_{average}(33) = \frac{35-33}{35-25} = \frac{2}{10} = 0.2$

"high" (30, 40, 40): $\mu_{high}(33) = \frac{33-30}{40-30} = \frac{3}{10} = 0.3$

Humidity = 65%:

"dry " (0, 0, 40): $\mu_{dry}(65) = 0$

" normal" (30, 50, 70): $\mu_{normal}(65) = \frac{70-65}{70-50} = \frac{5}{20} = 0.25$

"wet" (60, 100, 100): $\mu_{wet}(65) = \frac{65-60}{100-60} = \frac{5}{40} = 0.125$

$\alpha_1 = \min(0,0) = 0$

$\alpha_2 = \min(0.2, 0.25) = 0.2$

$\alpha_3 = \min(0.3, 0.125) = 0.125$

$\alpha_4 = \min(0.3, 0.25) = 0.25$

"average": $\max(0.2, 0.25) = 0.25$

"fast": 0.125

For each activated term, we calculate the center of gravity of the truncated triangle:
"average" (30, 50, 70):

$$S = \frac{1}{2} \cdot 40 \cdot 0.25 = 5$$

$$C = 50$$

"fast" (60, 100, 100):

$$S = \frac{1}{2} \cdot 40 \cdot 0.125 = 2.5$$

$$C = 100$$

$$FanSpeed = \frac{5 \cdot 50 + 2.5 \cdot 100}{5 + 2.5} = \frac{250 + 250}{7.5} = \frac{500}{7.5} \approx 66\ (\%)$$

6.For each parameter, find Q1 (25th percentile), Q3 (75th percentile) and IQR = Q3 – Q1. Compute the normal range boundaries:
lower = Q1 – 1.5·IQR,
upper = Q3 + 1.5·IQR.
Values outside the interval are considered anomalous.

Parameter	Q1	Q3	IQR	Normal range	Anomalies
Temperature	21	26	5	[13.5; 33.5]	Day 15 (45 °C)
Humidity	50.25	58	7.75	[38.625; 69.625]	Day 15 (15 %)

7.The output is a continuous numerical value (fan speed, %). The system predicts a specific number from input features (T, H), not a class label.

Answer: ***Correct answer:***

1.

$$\mu(x) = \begin{cases} 0, & \leq a \text{ or } x \geq c \\ \dfrac{x-a}{b-a}, & a < x \leq b \\ \dfrac{c-x}{c-b}, & b < x < c \end{cases}$$

2.

$\mu_{low}(33) = 0$

$\mu_{average}(33) = 0.2$

$\mu_{high}(33) = 0.3$

$\mu_{dry}(65) = 0$

$\mu_{normal}(65) = 0.25$

$\mu_{wet}(65) = 0.125$

3.

$\alpha_1 = 0$
$\alpha_2 = 0.2$
$\alpha_3 = 0.125$
$\alpha_4 = 0.25$

4.
"slow": 0
"average": 0.25
"fast": 0.125

5.
FanSpeed = 66%

6.
Anomalous values are detected on Day 15: temperature 45 °C and humidity 15 % — both exceed the acceptable IQR range

7.
Regression (with anomaly detection elements).

Assessment criteria

Criterion 1: The formula for the triangular membership function was written correctly — 0–1 points

Criterion 2: The degree of membership of input values to each term was calculated — 0–1 points

Criterion 3: Activation levels of output terms were determined for each of the k fuzzy rules — 0–2 points

Criterion 4: Output aggregation was carried out — 0–2 points

Criterion 5: Defuzzification was performed using the center of gravity method and a crisp FanSpeed value was correctly calculated as a percentage — 0–2 points

Criterion 6: Days with detected anomalies have been identified — 0–1 points

Criterion 7: The machine learning task type has been determined — 0–1 points

Task 44

For the oral-response section, the participant must choose one of the three tasks numbered 44, 45, and 46.

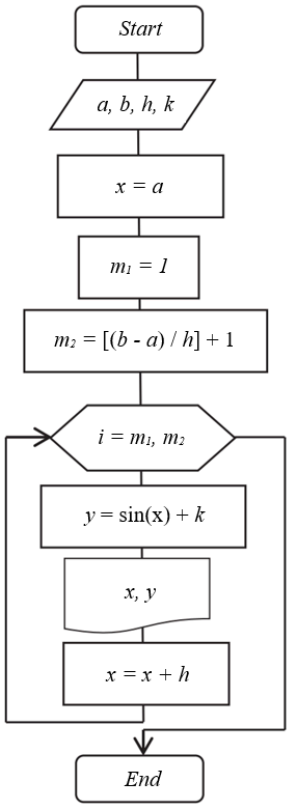
Oral-response task (maximum 5 points)

Field of science: Computer science, information systems

Advanced-level task: 41

It is necessary to introduce a check of the calculated value *y* against the condition of exceeding the given threshold *Y*threshold (*y*>*Y*threshold) into the algorithm shown in the figure. What data type is

advisable to choose for a flag variable that stores the indication of such an excess being detected? Indicate the physical meaning of the variable and the permissible set of values.



Answer: For the flag variable that stores the indication of threshold excess detection, the logical (Boolean) type (bool/boolean) should be chosen.
Physical meaning: the variable serves as a state indicator showing whether at least one value of *y* exceeding the threshold *Ythreshold* -has been detected during the loop execution. If an excess is detected, the flag is set to “true” (excess exists); otherwise, it remains “false” (no excess).
Permissible set of values: two mutually exclusive states — true or false.
Justification: using a logical type ensures semantic clarity of the code, as the variable directly reflects the result of a condition check. While a numeric type (int) storing 0 and 1 is technically possible, it reduces program readability and allows semantic ambiguity (values 0/1 may be interpreted differently), whereas the Boolean type is strictly limited to two states and is memory-optimal for storing event flags.

Assessment criteria

Criterion 1: Subject-specific knowledge (terminology, conceptual framework, and theoretical foundations).

Criterion 2: Breadth of knowledge (relevant facts, illustrative examples, and the integration of theory and practice).

Task 45

Oral-response task (maximum 5 points)

Field of science: Computer science, information systems

Advanced-level task: 42

In an array of measurement results, replace all negative values with 0 and round positive values to hundredths. Why use the ternary operator for this task? How can it be replaced?

Answer: It is not necessary to use a ternary operator here. But if it is used, its implementation has two goals: reducing the amount of syntax and increasing the semantic transparency of the codebase. You can use a regular if-else instead.

Assessment criteria

Criterion 1: Subject-specific knowledge (terminology, conceptual framework, and theoretical foundations).

Criterion 2: Breadth of knowledge (relevant facts, illustrative examples, and the integration of theory and practice).

Task 46

Oral-response task (maximum 5 points)

Field of science: Computer science, artificial intelligence

Advanced-level task: 43

Which two regression metrics are appropriate for evaluating the accuracy of the fuzzy fan speed model after removing anomalies using the IQR method?

Answer: MAE and RMSE, or MSE and R^2

Assessment criteria

Criterion 1: Subject-specific knowledge (terminology, conceptual framework, and theoretical foundations).

Criterion 2: Breadth of knowledge (relevant facts, illustrative examples, and the integration of theory and practice).